Ayoub Fourti

+216 55116920 ✉ fourtii.ayoub@gmail.com linkedin.com/in/ayoub-fourti github.com/ayoubfourti

Summary

Final-year engineering student (embedded systems, 5-year engineering degree) at ESPRIT, Tunis. **Robotics & control:** ROS 2, Nav2, SLAM (Cartographer), LiDAR/IMU/camera fusion, on-board YOLOv8 detection (Jetson Nano), sim-to-real (Gazebo), industrial AMR. **HMI & supervision:** Qt/QML, FastAPI, React, real-time monitoring. **Embedded:** STM32 bare-metal, ESP32, C/C++, Python, Linux.

Education

ESPRIT, School of Engineering 2022–present
Integrated preparatory cycle and engineering cycle *Ariana, Tunisia*

Experience

Cyber Mech Systems [View demo]
Software Engineering Intern — Robotics Platform *Remote, Summer 2026*

- Integrated a **ROS 2** stack on an **industrial AMR**: real-time control, autonomous navigation and mission supervision, validated in **Gazebo** before real-world deployment.
- Developed a fleet supervision HMI (FastAPI/React): real-time monitoring, telemetry and remote control; designed the architecture of a local Qt/QML HMI for the AMR (manual control, mission management, docking, diagnostics).

Academic Projects

Search-and-Rescue Hexapod Robot | *ROS 2, Jetson Nano, ESP32, Gazebo, SLAM, Nav2* [View demo]

- Implemented **SLAM (Cartographer)**, **Nav2** and **Behavior Trees** for autonomous navigation, with **LiDAR / IMU / camera** fusion and on-board **YOLOv8** detection on Jetson Nano.
- Developed **control and gait algorithms** for real-time control of **18 actuators** on **ESP32**: state estimation, phase-transition management and supervisory control.
- Validated the full stack in **Gazebo** (URDF model, contact and sensor plugins) before deployment on the real robot; identified and corrected **sim-to-real** gaps.

Smart AquaCare — Prototyping & Sensor Integration | *STM32, bare-metal C, ADC/DMA, PWM, IoT* [View demo]

- Developed a **real-time embedded controller** on **STM32** in bare-metal C: actuator control via **PWM** and **timers**, non-blocking sensor acquisition via **ADC + DMA**, interrupt-driven event handling.
- Automated scheduling with the **RTC** and deployed a **lightweight AI model** using **STM32Cube.AI**.

Fall-Detection Wristband (FPGA) | *VHDL, FPGA, Nios II, Quartus, ModelSim*

- Designed a modular **VHDL** fall-detection architecture for elderly people: **Moore state machine**, timer, threshold detection and counter, verified with a **6-scenario testbench** in **ModelSim** (team project).
- Implemented the same function through **hardware/software co-design** on **FPGA**: **Nios II** system in Platform Designer with application logic in C.

Wireless Auto — IoT Monitoring System | *ESP32, MQTT, Raspberry Pi* [View demo]

- Designed an **ESP32** sensor network for real-time monitoring, connected via **MQTT** to a **Raspberry Pi** gateway, an architecture close to field IoT monitoring solutions.
- Set up **remote supervision** through an IoT dashboard.

Skills

Programming: C/C++, Embedded C, Python, VHDL, Java, JavaScript, TypeScript, SQL, Bash
Embedded & firmware: Microcontrollers (STM32, ESP32, Arduino), DMA, timers, interrupts, UART, I2C, SPI, embedded Linux (Yocto, Buildroot)
FPGA & digital design: VHDL, Quartus Prime, ModelSim, Platform Designer, Nios II, HW/SW co-design
Robotics & control: ROS 2, computer vision (YOLOv8, OpenCV), on-board detection (Jetson Nano), real-time actuator control, STM32 bare-metal (PWM, timers, ADC/DMA), ESP32, Gazebo (sim-to-real), Nav2, multi-sensor fusion
HMI & supervision: Qt/QML, FastAPI, React, real-time monitoring, fleet supervision, multi-page HMI architecture
Tools: Git/GitHub, Docker, RViz, YOLOv8, AWS (certified), Proteus, VS Code
Languages: French (B2), English (B1)